Sequence Diagram Documentation

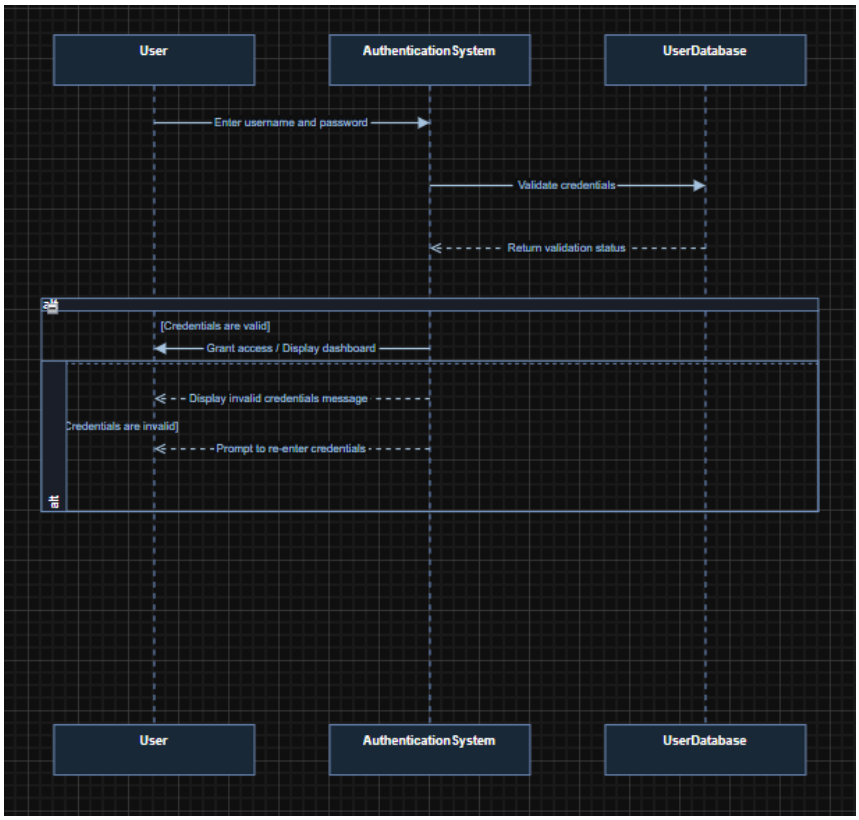
Student Academic Record Management System (SARMS)

A sequence diagram illustrates the dynamic behavior of a system by showing how objects interact with each other in a particular scenario of a use case. It represents the order of message exchange between actors and system components over time.

In the Student Academic Record Management System (SARMS), sequence diagrams are used to model interactions between actors such as **Administrator, Faculty, and Student** and system objects such as **Student, Course, Enrollment, Marks, and Report**.

Each sequence diagram corresponds to a specific use case identified in the system.

<u>1. Authenticate User Sequence Diagram</u>



Description

This sequence diagram illustrates how users authenticate themselves before accessing the system. Authentication is required for administrators, faculty, and students to ensure secure system access.

Actors and Lifelines - User (Administrator/Faculty/Student), SARMS System, Database

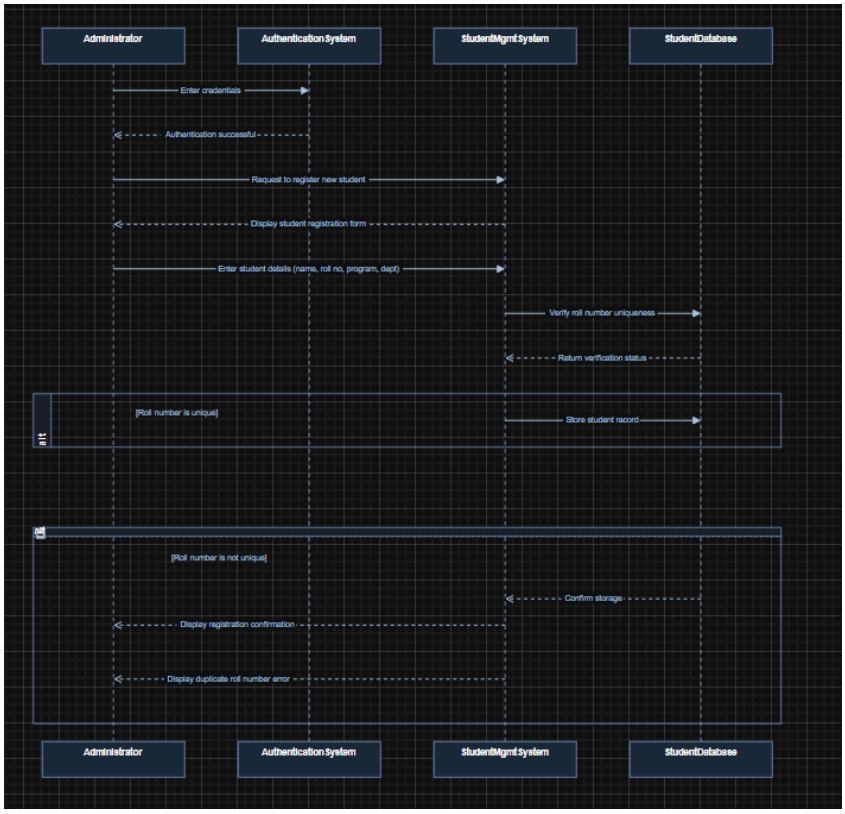
Sequence Flow

1. The user enters login credentials.
2. The SARMS system receives the login request.
3. The system sends a validation request to the database.
4. The database verifies the credentials.
5. If the credentials are valid, access is granted.
6. The system returns a successful login message to the user.

Alternative Flow

If the credentials are invalid, the system displays an authentication error message.

<u>2. Register Student Sequence Diagram</u>



Description

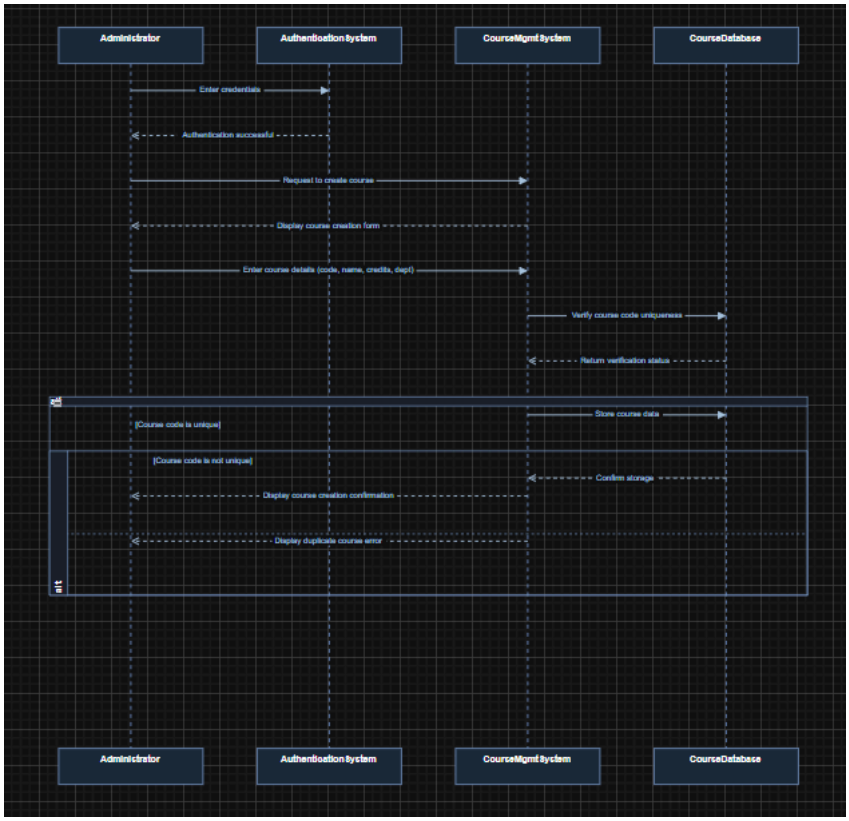
This sequence diagram shows how the administrator registers a new student in the system by entering personal and academic details.

Actors and Lifelines - Administrator, SARMS System, Student Object, Database

Sequence Flow

1. The administrator logs into the system.
2. The administrator submits student registration details.
3. The system creates a new student object.
4. A unique roll number is generated.
5. The student data is stored in the database.
6. The system confirms successful registration.

<u>3. Create Course Sequence Diagram</u>



Description

This sequence diagram represents the process of creating a new course by the administrator.

Actors and Lifelines - Administrator, SARMS System, Course Object, Database

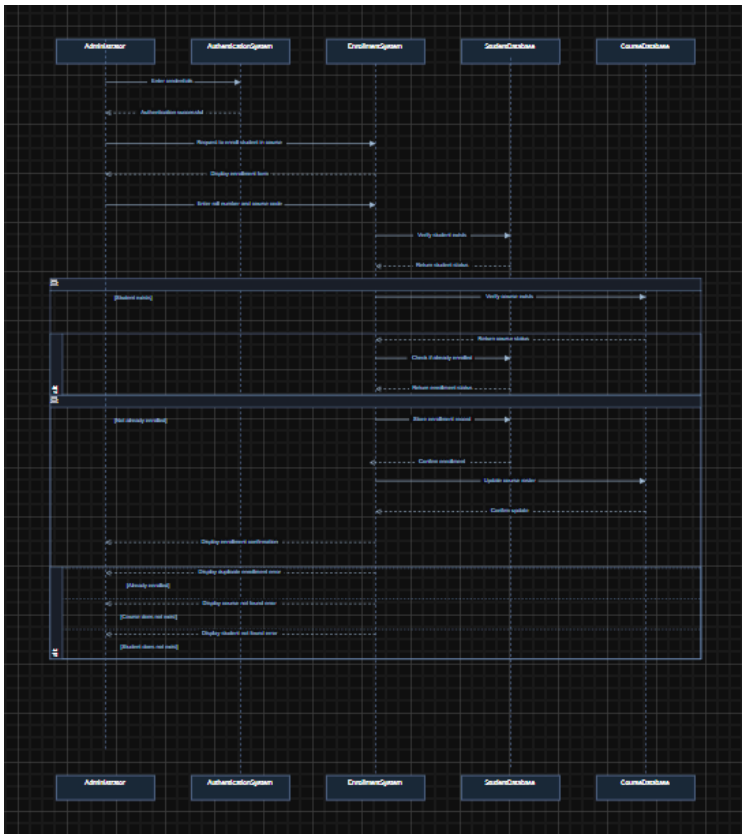
Sequence Flow

1. The administrator logs into the system.
2. The administrator submits course information such as course code, title, credits, and syllabus.
3. The system creates a course object.
4. The course information is stored in the database.
5. The system confirms successful course creation.

Alternative Flow

If the course already exists, the system displays an error message.

4. Course Enrollment Sequence Diagram



Description

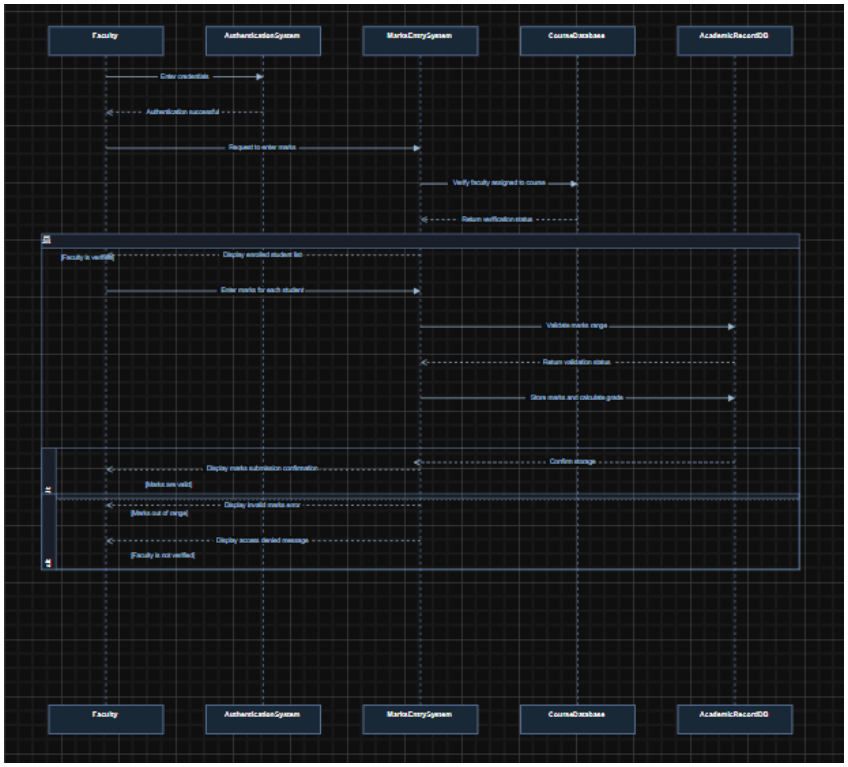
This sequence diagram describes how students are enrolled in courses by the administrator.

Actors and Lifelines - Administrator, SARMS System, Student Object, Course Object, Enrollment Object, Database

Sequence Flow

1. The administrator selects the student and course.
2. The system verifies that the student exists.
3. The system verifies that the course exists.
4. The system checks whether the student is already enrolled in the course.
5. A new enrollment record is created.
6. The enrollment information is stored in the database.
7. The system confirms successful enrollment.

5. Marks Entry Sequence Diagram



Description

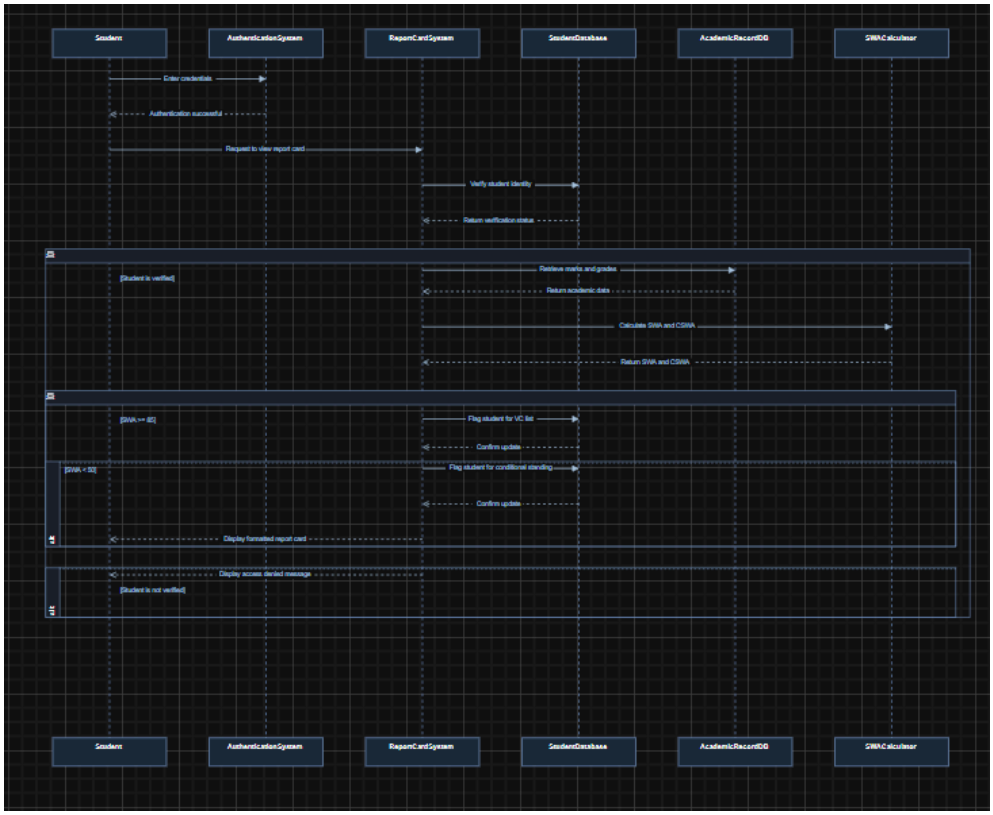
This sequence diagram shows how faculty members enter marks for students.

Actors and Lifelines - Faculty, SARMS System, Enrollment Object, Marks Object, Database

Sequence Flow

1. The faculty member logs into the system.
2. The faculty selects a course.
3. The system retrieves the list of enrolled students.
4. The faculty enters marks for each student.
5. The system stores the marks in the database.
6. The system calculates the grade for each student.

6. View Report Card Sequence Diagram



Description

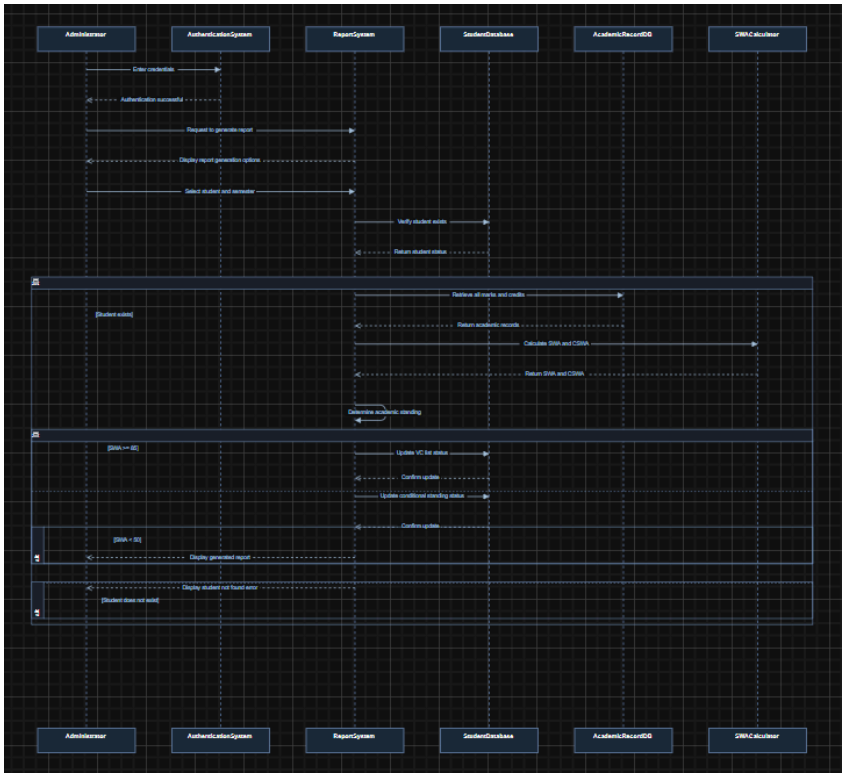
This sequence diagram illustrates how a student views their report card in the system.

Actors and Lifelines - Student, SARMS System, Enrollment Object, Marks Object, Report Object

Sequence Flow

1. The student logs into the system.
2. The student requests to view the report card.
3. The system retrieves enrollment details.
4. The system retrieves marks from the marks database.
5. The system calculates the Semester Weighted Average (SWA).
6. The system displays the report card.

7. Report Generation Sequence Diagram



Description

This sequence diagram shows how the system generates a complete academic report for a student.

Actors and Lifelines - Student, SARMS System, Marks Object, Report Object, Database

Sequence Flow

1. The student requests report generation.

2. The system retrieves all marks and enrollment information.
3. The system calculates the SWA and CSWA.
4. The system determines the student's academic standing.
5. A report object is generated.
6. The report is displayed or printed.